Forecasting Mental Health Distress in Online Forums Using Linguistic and Temporal Models
Final Reports

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Abstract

This study investigates early detection of psychological distress in online forums by integrating lexicon-based features (Empath and Circumplex valence-arousal mapping) with neural affect regressors derived from Mental-RoBERTa. Using 42,384 Beyond Blue posts and comments, we observe that over 80% of content clusters in negative, low-energy emotional quadrants, consistent with the forum's help-seeking context. Sentence-level valence–arousal regression achieves moderate alignment with lexicon targets (MAE 0.53; Pearson r 0.63), but exhibits variance compression on emotionally extreme posts.

To address sequence sparsity, we replace LSTM forecasting with a personalized ARX framework that fuses temporal signals, Circumplex trajectories, and RoBERTa-based affect predictions. The resulting multi-modal ARX model improves week ahead forecasting, reducing MAE to 0.32–0.33 and increasing directional accuracy relative to all single-modality baselines. SHAP analyzes show that temporal momentum is the dominant predictive factor, while linguistic and neural cues provide complementary refinements. The final system establishes an interpretable, data-efficient framework for modeling short-term emotional change in real-world mental-health forums.

1 Introduction

Mental health disorders represent a major global public health concern. In Australia alone, 18% of the population(approximately 5 million people) were dispensed mental health-related prescriptions(Australian Institute of Health and Welfare, 2023-2024). Although early intervention improves outcomes and reduces long-term social and economic burden, early detection is difficult because people often delay seeking help until symptoms become severe. Clinical assessments are also infrequent, resource-intensive, and inaccessible for many individuals.

In contrast, online mental-health forums provide a continuous stream of natural linguistic and emotional signals. Platforms like Beyond Blue allow users to express their feelings, describe symptoms, and seek support in real time, creating a rich source of information for studying emotional states and monitoring early states before formal diagnosis or clinical escalation. More and more studies demonstrate that linguistic patterns such as word choice, sentiment polarity, and thematic content can indicate changes in mental state. For example, Hur et al. (2022) [12] showed that weekly shifts in linguistic sentiment correlate with severity of depressive symptom, while Xu et al. (2024) [22] demonstrated that large language models (LLMs) can extract latent psychological cues in unstructured forum text. More recent work by Belcastro et al. (2025) [1] emphasizes the importance of explainability in mental-health prediction systems, especially to support clinical decision-making.

Despite these advances, many existing approaches still face several limitations. Most computational approaches rely on either linguistic analysis or LLM-based embeddings, without integrating multiple complementary signals. Many systems operate at the post level and overlook the temporal patterns that emerge weekly or monthly. Furthermore, although explainability has gained attention, neural models often function as black boxes. This limits clinical trust and practical implementation in sensitive mental-health contexts.

This research directly addresses these gaps by developing a multimodal framework that integrates linguistically grounded features (Empath categories [8]), affective representations (Circumplex valence–arousal coordinates[13]), and neural emotion estimates from Mental-RoBERTa[18]. Temporal modeling is then applied to forecast short-term changes in these trajectories. The goal is to provide early indicators of psychological distress that are both empirically grounded and transparent enough for real-world mental-health applications.

1.1 Objectives

- Build a structured dataset from Beyond Blue forums by web scraping, NLP preprocessing, and weekly user-level aggregation.
- Extract and integrate complementary forms of emotional information into a unified representation of weekly mental state.
- Develop and evaluate personalized forecasting models that predict short-term changes in users' emotional trajectories to support early identification of emerging distress.
- Improve interpretability of emotion forecasting through SHAP-based feature attribution, uncertainty diagnostics, and anomaly detection to ensure responsible use of predictive insights.

2 Background

Research on detecting mental-health signals from online text has expanded rapidly as large-scale digital traces have become easier to access. Linguistic behavior is widely recognized as a marker of psychological states, with several studies showing that shifts in language use mirror changes in emotional well being. Coppersmith et al. (2014) [6] demonstrated that depressive language patterns can be reliably extracted from social media, while Robertson et al. (2023) [16] showed that forum posts often reveal symptom progression. More recently, Hur et al. (2024) [12] reported that weekly variations in mood trajectories were associated with self-reported depressive symptoms, suggesting that temporal emotional patterns hold diagnostic relevance.

A central methodological debate in this area concerns the trade-off between interpretable psycholinguistic tools and more expressive but opaque neural models. Lexicon-based approaches such as LIWC [4] and Empath [8] offer transparency and have been used extensively to study depression, anxiety, and suicidality. However, these lexicons are not specifically designed for mental-health discourse and often broad or noisy categories. Cero et al. (2022) [5] pointed out that simple lexical counts frequently fail to capture subtle emotional difference, negation, sarcasm, or context, those limitations that are especially problematic in peer-support forums.

In contrast, large language models (LLMs) provide a deeper semantic understanding. Xu et al. (2024) [22] showed that domain-adapted RoBERTa models can infer latent psychological cues from unstructured forum text with high sensitivity, while Belcastro et al. (2025) [1] found that LLMs can extract clinically meaningful indicators from social-media narratives. However, these models present interpretability and reliability challenges, particularly in sensitive mental-health contexts. In additional, most LLM-based systems generate predictions at the post level, limiting their ability to represent longer-term behavioral patterns such as emotional trajectories spanning weeks or months.

Recent research increasingly supports modeling emotions in continuous affective spaces. The Circumplex Model of Affect (Russell, 1980) [13] provides the foundational valence–arousal framework, while recent extensions incorporate coping and regulation processes (Stanisławski et al., 2025) [19]. Computationally, text can be projected into this space using affective lexicons such as NRC-VAD (Mohammad, 2025 [15]) or neural regressors. Continuous emotional trajectories allow the study of volatility, drift, and recovery. These patterns are closely related to the progression of symptoms in clinical psychology. However, few computational studies integrate these emotional trajectories with psycholinguistic or LLM-derived features. This leaves a notable gap in unified multimodal modeling of emotional change.

Temporal forecasting of mental-health data remains underdeveloped. Deep sequence models such as LSTMs have been applied to emotional time-series when sufficient data are available (Hu et al., 2024 [10]). Extensions including Bayesian LSTMs for robustness (Wu et al., 2025 [21]) and attention-enhanced ConvLSTMs for sentiment analysis (Huang et al., 2021 [11]) show strong predictive performance. These models, however, often suffer from high data requirements and limited interpretability (issues that are magnified in sparse, user-generated forum data). In contrast, classical autoregressive models with exogenous inputs (ARX) remain widely used in econometrics and systems modeling for their transparency, coefficient interpretability, and stability under small data regimes (Box et al., 2016 [3]). Recent work even revisits ARX within neural identification frameworks (Dong et al., 2025[7]), highlighting its continued relevance. Despite this, ARX models have rarely been applied to mental-health forecasting, leaving an opportunity for interpretable temporal modeling.

The literature reveals several unresolved gaps. Psycholinguistic, affective, and neural features are typically studied recently in isolation despite each capturing complementary aspects of mental-health expression. Most computational work focuses on post-level classification rather than weekly emotional trajectories, even though temporal dynamics such as volatility and drift hold psychological significance.Interpretability remains a stated priority, yet many systems rely on opaque deep models and rarely incorporate explanation methods such as SHAP, LIME, or uncertainty-based diagnostics. Finally, temporal models that balance interpretability, personalization, and predictive stability remain largely unexplored, particularly for sparse, user-generated forum data.

2.1 Previous work in research Part A

In Part A of the project, we explored early detection of psychological distress in online forums using lexicon-based features (Empath categories and Circumplex valence–arousal coordinates) together with RoBERTa embeddings. A full scraping pipeline was implemented using Selenium and BeautifulSoup, covering 42,384 posts and comments from Beyond Blue across four mental-health domains.

Preliminary modeling in Part A highlighted several limitations. Dimensionality reduction preserved 76% of embedding variance, but downstream classifiers achieved only 31% accuracy and F1-scores below 0.20, showing that post-level features were too noisy and that CLS-pooled embeddings were unstable for short, informal texts. Moreover, the approach did not capture the weekly emotional dynamics that previous literature identifies as clinically informative.

These findings shaped the redesign in Part B. Word-level affect scores and CLS embeddings were replaced with sentence-level valence–arousal regression using Mental-RoBERTa, aggregated into weekly user trajectories. A multi-modal feature framework was introduced to integrate psycholinguistic cues, affective coordinates, and LLM regressions. LSTMs were evaluated but discarded due to sequence sparsity and unstable performance, motivating the adoption of a personalized ARX forecaster supported by SHAP explainability, uncertainty estimation, and anomaly analysis.

3 Methods

This study employs a multi-stage methodology designed to enable reproducible, interpretable early detection of mental health risks from large-scale online forum data.

3.1 Data Source and Acquisition

The dataset for this study was collected from Beyond Blue, a major Australian peer-support platform where users discuss mental-health challenges at different stages of severity. Four discussion boards were selected for their direct relevance to early risk detection: Anxiety, Depression, PTSD & Trauma, and Suicidal Thoughts & Self-Harm.

Data collection was performed using a custom Python pipeline integrated Selenium for rendering dynamically generated content and BeautifulSoup for structured HTML parsing. For each forum board, the scraper iterated up to 200 listing pages to capture thread-level metadata (title, category, URL, preview text). Individual threads were then opened and paginated to retrieve the main post and up to 40 comments to cover both short interactions and long-form discussions.

Temporal consistency was enforced by excluding entries without timestamps or dated earlier than 2019-01-01. A hybrid timestamp parser standardized dates in ISO-8601 format by handling both modern (span.local-date) and legacy (span.local-friendly-date[title]) formats. Each retrieved item contained the following fields: post/comment text, author identifier, normalized timestamp, category, thread URL, and a structured list of associated comments.

To support user-level temporal analysis, posts and comments were linked using thread IDs and pseudo-anonymous user identifiers. JSON-encoded comment blocks were flattened so that each message (post or comment) became a single analytical unit. The final corpus contained 42,412 text entries (6,616 main posts and 35,796 comments) spanning 2019-01-01 to 2025-07-18.

3.1.1 Ethical Considerations and Data Governance

All data were collected solely from publicly accessible Beyond Blue forum pages. No private or restricted content was accessed. During preprocessing, usernames, URLs, and other identifiable strings were removed, and all user IDs were replaced with random hashes to prevent re-identification. No demographic information was inferred or reconstructed. Scraping respected rate-limits and avoided disruptive actions. The study followed the University of Adelaide's ethical guidelines for research with publicly available online data and maintained strict privacy protection throughout.

3.2 Text Preprocessing

All scraped forum content underwent a standard NLP step to convert all sentences into structured input for downstream feature extraction. Duplicated posts and repeated content generated through quoting or thread pagination were removed to prevent over-representation of single users. Text was then normalized by lowercasing, removing HTML tags, markup symbols, URLs, email-like strings, and JavaScript residue.

Given the importance of preserving linguistic markers relevant to mental-health expression, stop-words, pronouns, and negations were intentionally retained. Instead, light normalization steps were applied, including standardizing whitespace, reducing elongated characters (e.g., "sooo"), and collapsing repeated punctuation. Non-semantic artifacts (emoji placeholders, broken characters) were removed only when they lacked affective content.

All timestamps were converted to ISO-8601, and each text entry was linked to a hashed user identifier to support longitudinal aggregation at the user-week level. The final cleaned text field served as input to two feature extraction streams:

- lexicon-based extraction (Empath categories, NRC-VAD valence/arousal scores), and
- LLM-based embedding and regression models (Mental RoBERTa).

3.3 Psycholinguistic features

Empath [8] was used to extract interpretable psycholinguistic features from the cleaned text. The lexicon provides 194 categories that include emotional expression, cognition, interpersonal behavior, and everyday themes. Although not specifically designed for clinical analysis, Empath-derived topic patterns have been shown to correlate with population-level mental distress (Bowen et al., 2020 [2]) and remain effective in detecting psychological signals in online discourse (Clavy et al., 2023 [9]). These properties make Empath an appropriate complement to the neural representations used in this multi-modal framework.

For each post or comment, Empath generated proportional category scores normalized by text length. These document-level features were then aggregated into weekly user profiles to match with temporal resolution used in Circumplex valence–arousal estimation and Mental-RoBERTa regression outputs. Weekly aggregation allows linguistic behavior to be treated as a trajectory rather than a collection of isolated posts.

Because Empath contains a broad set of general-purpose categories, feature selection was required to retain only dimensions relevant to mental-health expression. Categories showing minimal variation in the corpus or lacking psychological interpretability were removed. In contrast, categories related to affective tone, interpersonal support, cognitive strain, and help-seeking behaviors were retrained. A small number of mental-health-specific composite categories were also added to improve representational relevance. Categories with high instability ($CV > 0.5$) were excluded to avoid noise from propagating into trajectory-based forecasting. The resulting set of 28 categories formed the final subset of Empath of project subset.

Within the multi-modal feature architecture, Empath contributes a transparent linguistic signal that complements the more abstract representations from Mental-RoBERTa and the continuous affective space constructed through the Circumplex model. In the ARX forecasting stage, several Empath categories provided measurable improvements in short-term emotional prediction, and their explicit semantic meaning supports downstream SHAP analyzes by linking attributions to identifiable linguistic behaviors.

3.4 Valence–Arousal Modeling

To obtain continuous affective representations of user emotional states, all cleaned text was projected into the valence–arousal space of the Circumplex Model of Affect (Russell, 1980 [13]; Stanisławski et al., 2025 [19]). The model's two orthogonal dimensions—valence (pleasant–unpleasant) and arousal (activated–deactivated)—provide a compact representation is suitable for tracking emotional dynamics over time.

3.4.1 Document-Level Affective Representation

Each post and comment was mapped in [0,1] using the NRC-VAD v2 lexicon (Mohammad, 2025 [15]). Tokens with VAD entries contributed to document-level affect, computed as:

$$v_d = \frac{1}{|W_d|} \sum_{w \in W_d} VAD_v(w), a_d = \frac{1}{|W_d|} \sum_{w \in W_d} VAD_a(w) \tag{1}$$

where W_d is the set of lexicon-covered tokens in document d. This produced a continuous valence–arousal coordinate for each document.

Figure 1 visualizes the full distribution of document-level pairs, revealing a strong concentration in negative and low-energy regions—consistent with the urgent, distress-oriented nature of help-seeking forums.

3.4.2 Aligning Linguistic Categories with Affective Space

To provide a psychologically interpretable orientation for the Circumplex model, we projected Empath categories into the same valence–arousal plane. For each category, valence–arousal coordinates were calculated as the weighted average of VAD scores across all documents where the category appeared.

We also computed affective vectors for Empath categories by averaging the VAD scores of all documents containing each category (Figure 2). Positive-affect categories (optimism, affection) cluster in the upper-right quadrant, while distress-related categories (sadness, depression) concentrate in the lower-left, aligning with psychological theory and validating the lexicon–affect linkage.

3.4.3 Stabilizing Affective Geometry (Whitening)

Although z-scoring standardizes scale, valence and arousal remain moderately correlated in natural language. To guaranty that distances and

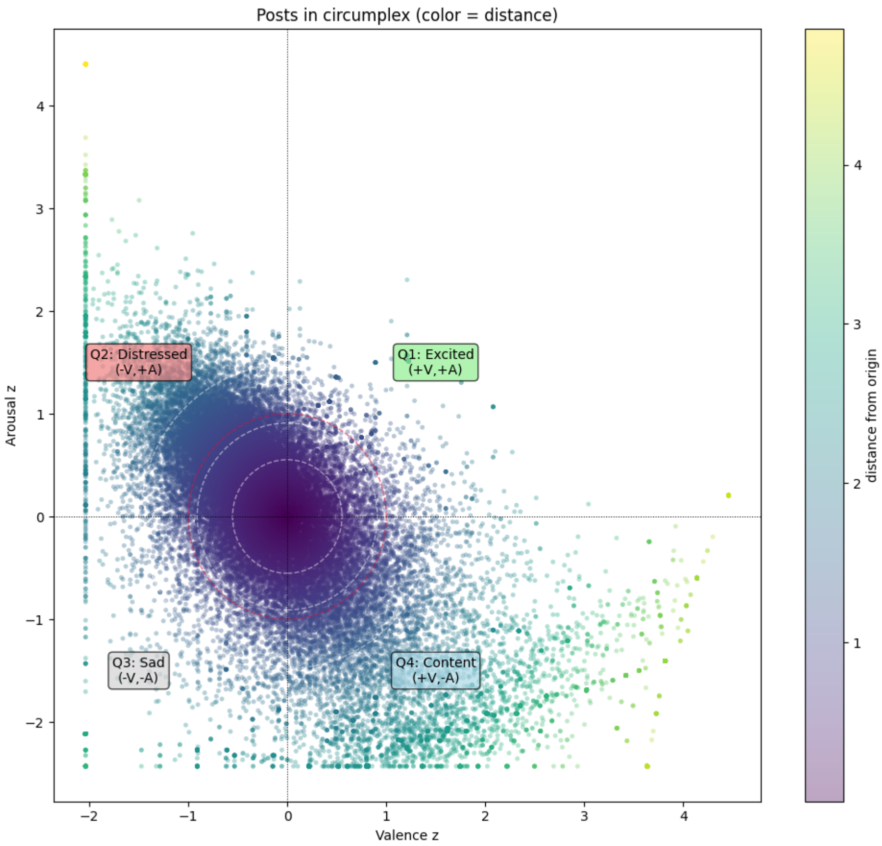

Figure 1: Distribution of z-scored valence and arousal

angles in the affective plane capture genuine emotional variation rather than linguistic covariance, we applied a whitening transform. Whitening decorrelates features and produces an isotropic geometry [14] to improve stability for linear models and the interpretability of directional analyzes. Recent work shows that whitening reshapes optimization geometry into a more stable and better-conditioned space [17].

The density plot in Figure 3 shows that post-whitening distributions become radially symmetric, supporting the ARX model's independence assumptions and enabling a reliable angle-based emotional drift analysis.

3.4.4 Temporal Aggregation into Weekly Trajectories

Document-level valence and arousal values were aggregated into user-week trajectories using simple averaging:

$$v_{u,w} = \frac{1}{|D_{u,w}|} \sum_{d \in D_{u,w}} v_d, \, a_{u,w} = \frac{1}{|D_{u,w}|} \sum_{d \in D_{u,w}} a_d \quad (2)$$

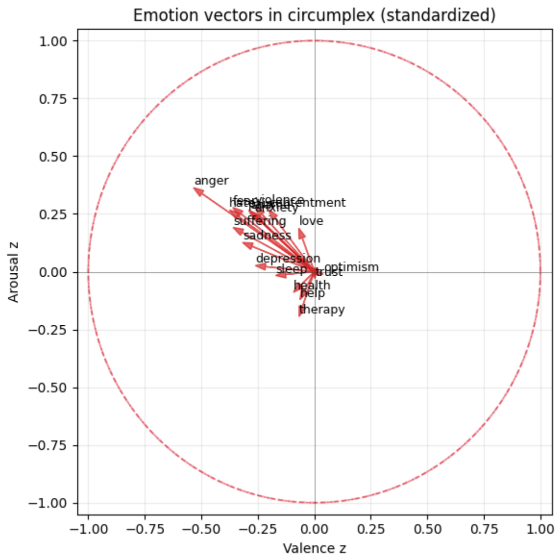

Figure 2: Empath category vectors in z-scored valence–arousal space

where $D_{u,w}$ is the set of documents authored by user u in week w. Weekly z-scoring produced user-normalized trajectories that serve as both (i) forecasting targets and (ii) exogenous predictors in the ARX model.

We additionally derived five auxiliary symptom-prototype scores (anxiety, depression, PTSD-like distress, manic activation, suicidality) using Gaussian similarity to affective prototypes combined with z-scored Empath signals (Appendix A.1). These trajectories offer interpretable behavioral markers that can be used as optional exogenous covariates in forecasting experiments.

3.5 Neural Modeling with Mental-RoBERTa

To obtain richer psychological signals than lexicon-based features alone, we employed Mental-RoBERTa, a RoBERTa based transformer pre-trained on large mental-health corpora [18]. Prior studies have shown that domain-adapted transformers capture symptom language, coping expressions, and informal forum discourse more reliably than general-purpose BERT models [20].

In Part A of the project, we used a CLS embedding as a static feature, but performance was weak and the representation was difficult to interpret. In this stage, the model was reframed as a continuous valence–arousal regressor to make consistency with the Circumplex framework and to make a comparison with lexicon-derived affect.

Regression Architecture

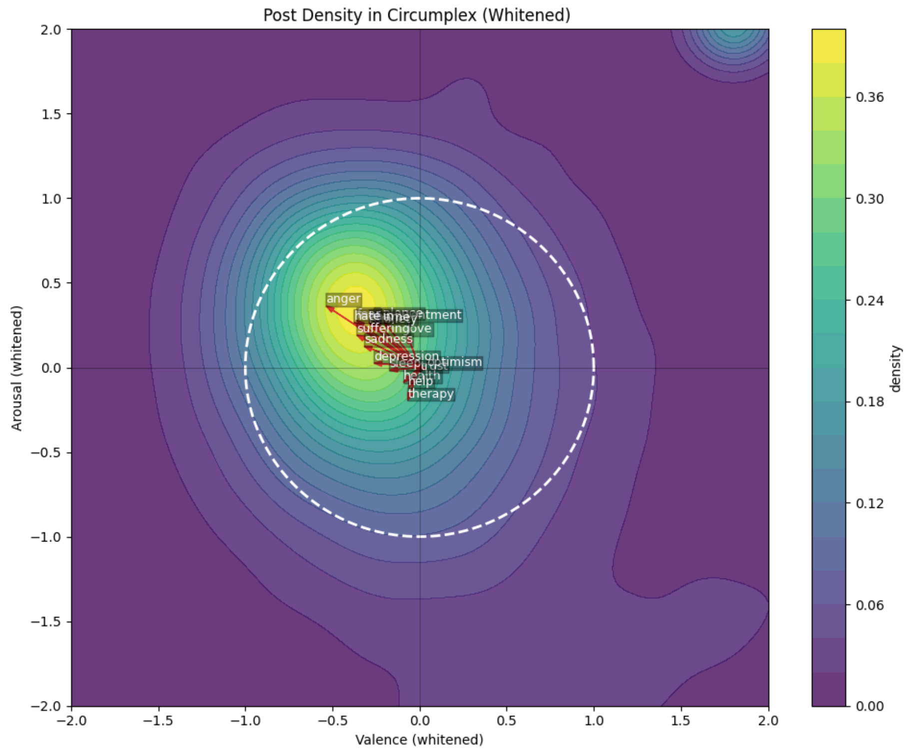

Figure 3: Kernel density estimate of whitened valence–arousal

Each cleaned post or comment is tokenized and encoded using the frozen Mental-RoBERTa transformer. Contextual token embeddings are mean-pooled into a sentence vector and passed through a two-layer regression head that outputs predicted valence $\hat{v}_d$ and arousal $\hat{a}_d$ for document d. The head is trained to match NRC-VAD ground truth via mean-squared error:

$$L_{MSE} = \frac{1}{N}\sum_{d=1}^{N}[(\hat{v}_d - v_d)^2 + (\hat{a}_d - a_d)^2] \tag{3}$$

Model selection uses validation MAE and Pearson correlation (details in Section 4). This design treats emotion as a continuous variable rather than a categorical label and ties transformer predictions directly to the same affective geometry used in Circumplex.

Weekly Aggregation After training, the regressor is applied to all 42k+ documents. Predictions are aggregated into user-week trajectories by averaging across all posts/comments authored in each week. These trajectories serve two key purposes:

1. **Independent affect estimate**They provide a contextual signal

that can confirm or diverge from lexicon-based valence–arousal patterns, especially when lexical cues are sparse.

2. **Exogenous predictors for forecasting** Weekly predicted valence–arousal coordinates are incorporated into the ARX model to quantify how shifts in transformer-inferred affect contribute to short-term emotional change.

Rationale This restructuring resolves limitations from the earlier phase of the project. By grounding the model in the Circumplex space, the neural outputs become interpretable and psychologically meaningful. By aggregating predictions to the weekly level, the model integrates naturally into temporal forecasting and SHAP-based attribution rather than remaining an opaque high-dimensional vector.

3.6 Temporal Forecasting

Detecting early signs of distress accurately requires modeling how users' emotional states evolve over time. Peer-support forums activity are highly irregular(users post intermittently, with variable text volume and long gaps between updates) making deep sequence models difficult to train reliably. Initial LSTM experiments confirmed this: gradients were unstable, predictions collapsed toward the mean, and hidden-state representations offered limited interpretability. These limitations motivated a shift toward interpretable linear dynamics better suited to sparse behavioral data.

Thus, we employed an AutoRegressive model with eXogenous inputs (ARX) to forecast weekly valence or arousal. For each user, the emotional state at week t is modeled as a function of (i) their own recent history and (ii) multimodal linguistic signals extracted in earlier stages:

$$y_t = \sum_{i=1}^{p} a_i y_{t-i} + \sum_{j=1}^{q} b_j x_{t-j} + \epsilon_t \tag{4}$$

where y_t denotes valence or arousal, x_{t-j} represents exogenous predictors (Empath categories, whitened Circumplex trajectories, Mental-RoBERTa regressions, and optional symptom indicators), a_i, b_j are learned coefficients, and ϵ_t is the residual error.

This structure mirrors linguistic and emotional patterns observed in one week and should influence near-term shifts in mood or distress.

ARX offers several advantages under real forum conditions. First, it handles sparse and unevenly spaced sequences through explicit temporal indexing, avoiding the instability encountered in recurrent networks.

Second, it provides transparent linear coefficients that are direct interpretation of how linguistic cues contribute to emotional change. This interpretability is crucial for clinical or behavioral research contexts where opaque neural states provide limited infomation.

Model performance is evaluated using weekly MAE and Pearson correlation between predicted and observed valence/arousal, consistent with earlier modules. Since ARX is linear with well-defined residual distributions, it also supports downstream uncertainty quantification and anomaly detection—tools later used to assess prediction reliability and identify emotionally volatile periods (Section 4).

In this framework, ARX serves as the core temporal model linking multi-modal weekly signals (psycholinguistic features, continuous affective coordinates, and transformer-based emotion regressions) to short-term shifts in user mood.

3.7 Model Explainability

Because the ARX forecasting model integrates multiple heterogeneous signals (weekly valence–arousal trajectories, Mental-RoBERTa regressions, Empath categories, and symptom prototypes) an explicit explanation framework was required to interpret how these inputs contributed to predicted emotional changes. To maintain transparency, we applied SHAP values, which are well-suited to ARX since the model is linear and additive. In this setting, SHAP produces exact analytic attributions rather than approximations, allowing each weekly forecast to be decomposed into contributions from individual features.

For a given user u, the ARX prediction for week t can be expressed as:

$$\hat{y}_{u,t} = \theta_{u,0} + \sum_{l=1}^{p} \theta_{u,l}^{(y)} y_{u,t-l} + \sum_{j=1}^{q} \sum_{k=1}^{K} \theta_{u,j,k}^{(x)} x_{u,t-j,k} \tag{5}$$

Since this structure is strictly additive, each lagged emotional term and each exogenous feature contributes a fixed, signed amount to the prediction. SHAP assigns a value:

$$\phi_{u,t,k} = \theta_{u,j,k}^{(x)} x_{u,t-j,k} \tag{6}$$

and the attributions sum exactly to the offset from the user's baseline:

$$\sum_{k} \phi_{u,t,k} = \hat{y}_{u,t} - E[y_u] \tag{7}$$

This provides a transparent decomposition of why the model predicts an increase or decrease in valence or arousal in a specific week.

SHAP helps to clarify how the different parts of our model contribute to each prediction. Our features often move in different ways: the Circumplex trends capture slow changes in mood, RoBERTa captures moment-to-moment language signals, and symptom scores highlight clinically meaningful spikes. Instead of guessing which signal caused a shift, it clears showing (eg. whether a drop in valence was driven by symptom scores or simply by the user's recent emotional trajectory).

It also reveals how the ARX model behaves for each person. Some users are primarily governed by their own past emotional states, while others are more influenced by linguistic cues such as fear, sadness, or sleep-related language. Seeing these differences helps avoid misleading generalizations based on population averages. When a prediction error occurs, we can see whether it reflects a true emotional jump or uncertainty caused by sparse posting or inconsistent text. Because SHAP follows the structure of the ARX model, the explanations stay grounded and reproducible.

3.8 Evaluation Metrics

A unified evaluation framework was required to assess the different components of this study (lexicon features, neural regressors, symptom scores, and the temporal forecasting model). The chosen metrics emphasize accuracy, directional reliability, temporal consistency, and interpretability, reflecting the needs of an early distress detection setting.

For continuous emotion regression (Mental-RoBERTa and ARX), the primary accuracy measure is Mean Absolute Error (MAE):

$$\mathrm{MAE} = \frac{1}{N}\sum_{t=1}^{N} |\hat{y}_t - y_t| \tag{8}$$

MAE is widely used in affective computing because it directly quantifies the typical deviation between predicted and reference valence–arousal scores. To complement MAE, we compute the Pearson correlation coefficient, which evaluates whether the model captures the direction and shape of emotional trajectories:

$$r = \frac{\sum_{t=1}^{N}(\hat{y}_t - \bar{\hat{y}})(y_t - \bar{y})}{\sqrt{\sum_{t=1}^{N}(\hat{y}_t - \bar{\hat{y}})^2}\sqrt{\sum_{t=1}^{N}(y_t - \bar{y})^2}} \tag{9}$$

Correlation is particularly important because many downstream analyzes (e.g., volatility estimation, anomaly detection) depend on predicting whether affect is rising or falling rather than its absolute value.

For temporal forecasting, we also evaluate Directional Accuracy (DirAcc), which measures whether the model correctly predicts emotional improvement or deterioration:

$$\text{DirAcc} = \frac{1}{N}\sum_{t=1}^{N} \mathbf{1}\Big[\text{sign}(\hat{y}_t - \hat{y}_{t-1}) = \text{sign}(y_t - y_{t-1})\Big] \tag{10}$$

This metric aligns with early-warning objectives, where detecting week-to-week increases in distress is more important than precise numerical forecasts.

For Empath features, we assessed temporal stability using the Coefficient of Variation (CV):

$$CV_k = \frac{\sigma_m(\bar{e}_{k,m})}{\mu_m(\bar{e}_{k,m})} \tag{11}$$

Categories with high instability were removed to prevent noise from entering weekly forecasting models.

For the ARX component, model diagnostics included checks for residual independence, constant variance, and absence of autocorrelation. These tests confirm that remaining error does not contain an unmodeled temporal structure.

Finally, model interpretability was evaluated through SHAP values, which provide an exact decomposition of ARX predictions into contributions from individual features. SHAP does not measure numerical accuracy but ensures that predictions can be traced to specific linguistic or affective signals as an essential requirement in mental-health forecasting.

4 Results

4.1 Component-Level Results

Circumplex Emotion Signals To characterize the affective feature of the Beyond Blue forum, all posts and comments were projected into the continuous valence–arousal (VA) space using NRC-VAD scores. The raw distributions show a pronounced negativity bias consistent with the help-seeking context of the platform. In 42,384 documents, mean valence was –0.375 and mean arousal –0.384, with most values lying between –1.0 and –0.2 on both axes. As shown in Table 1, 85.7% of all documents fall in the low-valence, low-arousal quadrant, which means the majority of forum content reflects subdued and unpleasant affect.

Because raw VA scores are influenced by global lexical frequency patterns, all values were standardized (z-scored) prior to downstream modeling. As expected, z-scoring re-centers the distribution at zero (valence_z

Emotion Quadrant	**Quadrant**	**Count**	**Percent**
Low Negative (Sad)	Q3	36,333	85.7%
Low Positive (Content)	Q4	3,670	8.7%
High Positive (Excited)	Q1	1,391	3.3%
High Negative (Distressed)	Q2	990	2.3%
Total		42,384	100%

Table 1: Raw Emotion Quadrant

= –0.000, arousal_z = 0.000) and imposes unit variance. This transformation also alters quadrant proportions, since the neutral point becomes the dataset mean rather than the midpoint of the NRC-VAD scale. After z-scoring (Table 2), the distribution becomes more balanced, with 44.6% of posts lying in Q2 (negative valence, high arousal) and 27.6% in Q4.

Emotion Quadrant	**Quadrant**	**Count**	**Percent**
High Negative (Distressed)	Q2	18,919	44.6%
Low Positive (Content)	Q4	11,718	27.6%
Low Negative (Sad)	Q3	6,492	15.3%
High Positive (Excited)	Q1	5,255	12.4%
Total		42,384	100%

Table 2: Z-score Valence/Arousal

Although Table 2 indicates that 12–13(i) few documents exceed a raw arousal of +0.3, and (ii) $> 40,000$ points overlap densely near the bottom-left region, visually masking minority quadrants. Thus, the visual sparsity in Q1/Q2 reflects over-plotting, not a computational error.

Figure 4 shows the raw VA distribution, highlighting the natural emotional skew. Figure 1 displays the z-scored distribution used for forecasting. Together, these plots confirm that emotional expression on the forum is overwhelmingly negative and low-energy, and that standardization preserves structural patterns while ensuring compatibility with Empath and neural regressors in the ARX stage.

Figure 4: Raw Valence/Arousal Distribution

Mental-RoBERTa Regression Outputs To complement lexicon-derived affect signals, we evaluated Mental-RoBERTa as a continuous valence–arousal regressor. On the held-out test set, the model achieved MAE = 0.5295 (valence) and 0.5236 (arousal), with Pearson correlations of 0.633 and 0.639 and Spearman coefficients of 0.640 and 0.693, respectively (Table 3). These metrics indicate that the model reliably captures the direction and relative ordering of emotional variation, even though post-level affect estimation remains challenging due to short, conversational forum text.

Calibration analysis reveals slopes of 0.436 (valence) and 0.438 (arousal), both substantially below 1, showing that predictions are variance-compressed (extreme emotions tend to be underestimated). However, the intercepts remain small (0.04 and 0.01), indicating no meaningful global bias. The hexbin plots in Figure 5 illustrate this pattern: points cluster along the diagonal but remain confined to narrower bands than the ground-truth range.

Residual diagnostics support with these observations. As shown in Figure 7, valence residuals center near zero (mean 0.035, SD 0.809), while arousal residuals exhibit a similar distribution (mean 0.020, SD 0.796). Both histograms are approximately symmetric with moderately

V/A	MAE	Pearson r	Spearman p	Slope	Intercept
Valence	0.5295	0.633	0.640	0.436	0.039
Arousal	0.5236	0.639	0.693	0.438	0.013

Table 3: Overall Regression Performance

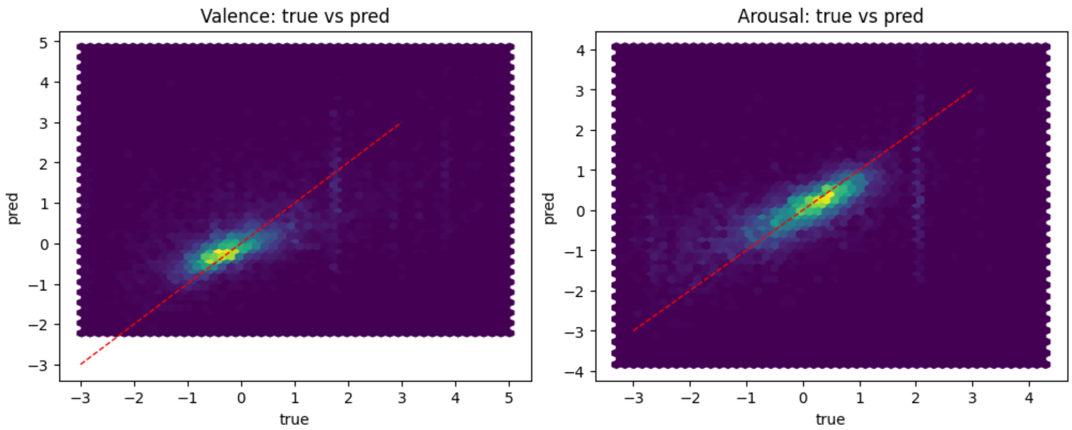

Figure 5: Hexbin Density

heavy tails, reflecting stable average behavior but occasional larger errors for extremely short or emotionally ambiguous posts. These patterns are expected in lexicon-supervised fine-tuning, where the target signal is inherently noisy.

Model sensitivity to text length is shown in Table 4 and Figure 6. Error declines monotonically with increasing document length, showing that richer linguistic context produces more reliable affect estimates. Very short posts (common in crisis-driven threads) offer limited lexical cues, increasing prediction variance.

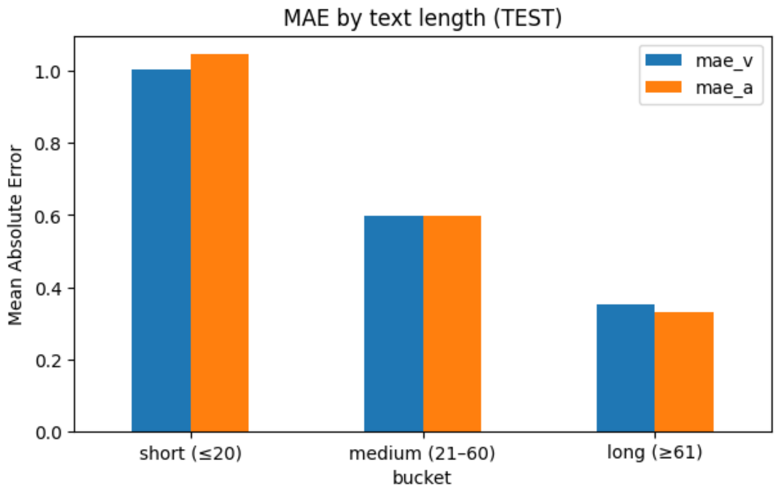

Figure 6: MAE text length

Length bucket	**MAE(valence)**	**MAE(arousal)**
short (≤ 20)	1.003	1.046
medium (21 60)	0.598	0.597
long (≥ 61)	0.353	0.331

Table 4: Length-bucket MAE

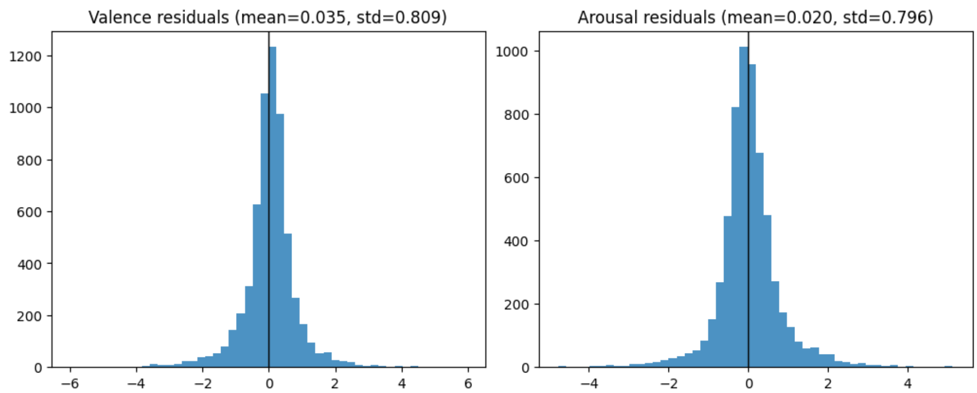

Figure 7: Residual Histogram

In general, Mental-RoBERTa provides a useful but imperfect affective signal. It captures a broad emotional direction with moderate precision and integrates naturally with the Circumplex coordinates through its shared valence–arousal space. While the model cannot resolve all nuances of forum discourse, its structured weekly estimates contribute meaningful exogenous information for the ARX forecasting stage.

4.2 Forecasting outputs

We first assessed whether deep sequence models could forecast week-to-week emotional changes. A univariate LSTM was trained on user-level valence–arousal trajectories, but performance was poor. Directional accuracy reached only 53.2% for valence and 50.0% for arousal, which is essentially equivalent to random guessing. More critically, the model suffered from severe variance collapse: predicted changes had an extremely narrow spread (both SD 0.03) compared to the true variability (1.03 for both). Nearly 70% of valence predictions were effectively zero, showing that the model defaulted to flat trajectories rather than learning meaningful temporal patterns. These failure modes (well-documented in sparse behavioral sequences) motivated the transition to a more stable and interpretable ARX framework.

The ARX models showed substantially better performance and consistency among users. Table 5 summarizes results in all feature families. Using only Circumplex inputs produced MAE = 0.351 and directional accuracy = 0.619 for valence; Mental-RoBERTa features performed similarly (MAE 0.347). Symptom-prototype features achieved comparable accuracy, confirming that each modality captures partly overlapping affective signals. The gains become more pronounced when modalities are combined. The fusion model (Circumplex + RoBERTa) reduced valence MAE to 0.352 and improved directional accuracy to 0.649, while the fusion + symptoms model achieved the best results overall: MAE_V = 0.322, MAE_A = 0.326, with directional accuracy = 0.652 and 0.644. Although absolute improvements are modest (3–5% MAE reduction), they are consistent across users and across both affect dimensions.

Feature Set	**MAE_V**	**MAE_A**	**DirAcc_V**	**DirAcc_A**	**n_feats**
Circumplex	0.3507	0.3576	0.619	0.609	14
Mental-RoBERTa	0.3466	0.3520	0.623	0.612	14
Symptom Features	0.3478	0.3533	0.624	0.615	14
Fusion	0.3523	0.3313	0.649	0.637	38
Fusion + Symptoms	0.3219	0.3264	0.652	0.644	43

Table 5: ARX forecasting performance across feature sets

These trends are visible in the Feature Set Comparison bar chart (Figure 8), where fusion-based models outperform single-source baselines. Importantly, the advantage of fusion persists even when user-level variability is considered. The per-user rolling MAE distribution (Figure 9) shows a median error near 0.50, but with large variance across individuals: stable users exhibit predictable trajectories, whereas users with irregular or abrupt emotional changes remain challenging. This mirrors earlier descriptive analysis of volatility in Section 4.1 and further illustrates why forecasting accuracy varies strongly by user.

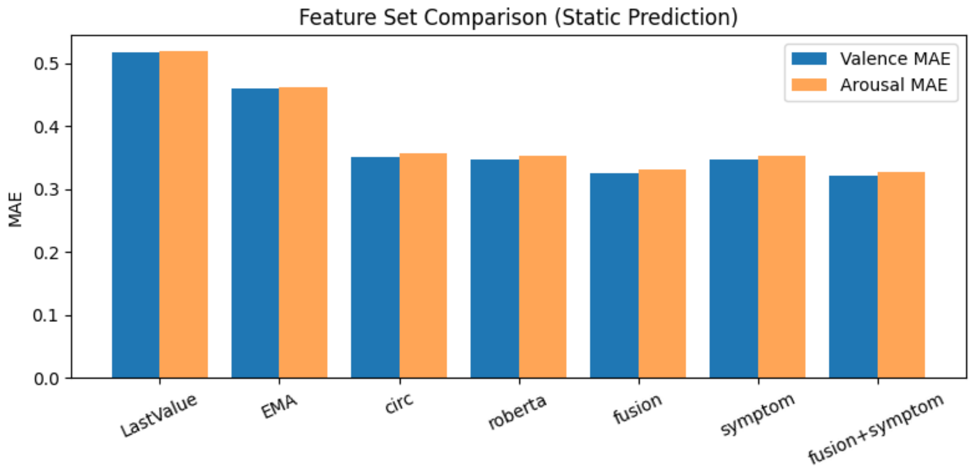

Figure 8: Feature Set Comparison

Feature-level interpretability further clarifies these patterns. Permutation-importance analysis (Figure 10) identifies the most influential predictors of valence: recent temporal momentum (e.g., rm2_slope3), short-term variability (rm2_std4), and selected symptom-aligned linguistic categories (help, therapy, anger). Both Circumplex and RoBERTa features appear among the top contributors, indicating that the ARX model genuinely exploits complementary sources of information rather than over-relying on any single feature family.

The ARX approach delivers moderately accurate, directionally reliable, and interpretable forecasts of emotional change. It avoids the instability of LSTMs, integrates multimodal signals in a transparent linear structure, and provides a foundation for the interpretability and uncertainty analyzes that follow in Section 4.3.

4.3 Model Interpretability

SHAP analysis was conducted on the fusion ARX model (Circumplex + Mental-RoBERTa + temporal features). Although the fusion + symptom configuration produced slightly stronger forecasting accuracy, in-

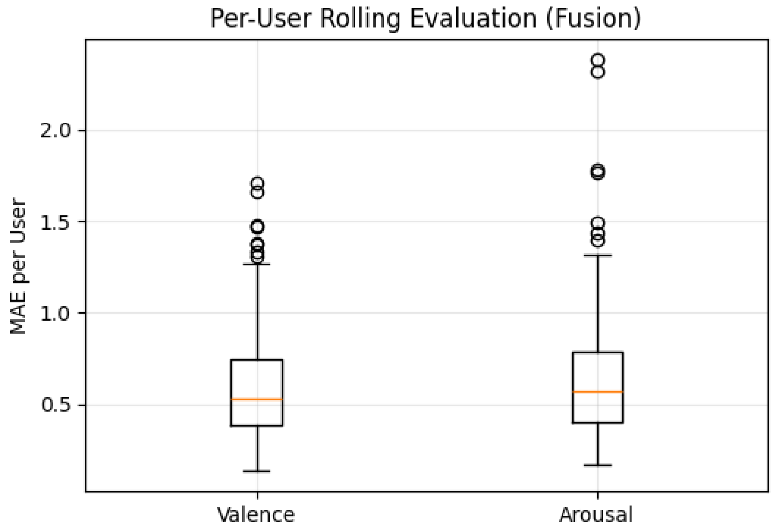

Figure 9: Per-User Rolling Evaluation

terpretability analzes were finalized before symptom scores were integrated. More importantly, the fusion model retains the core behavioral and affective signals (recent valence–arousal history, RoBERTa regression outputs, and Empath-derived patterns)that dominate all ARX variants. The SHAP findings therefore remain representative of the entire forecasting pipeline. Valence explanations are presented here; the complete valence/arousal set appears in Appendix A.2.

Global Feature Contributions Global SHAP results (Figure 13) reveal a clear hierarchy between predictors. Temporal dynamics overwhelmingly drive the model's behavior. The most influential feature is the exponentially smoothed valence trend, v_true_mean_rm2_ema3, which consistently exhibits the largest attribution magnitudes across users. The next strongest contributor is v_last, indicating that the model heavily base on the most recent emotional state when estimating next-week valence. These two features alone explain a substantial proportion of predictive variation.

Linguistic variables contribute only modestly. A small number of Empath categories (such as shame_mean_z and pain_mean_z) appear in the global ranking but with substantially smaller SHAP magnitudes, reflecting their role as secondary cues rather than primary temporal drivers. Circumplex-derived engineered features also contribute at low but consistent levels, while behavioral variables such as posting frequency and volatility have almost no influence, aligning with the ablation findings in Section 4.3.3.

The distribution of mean absolute SHAP values (Figure 15) quantifies

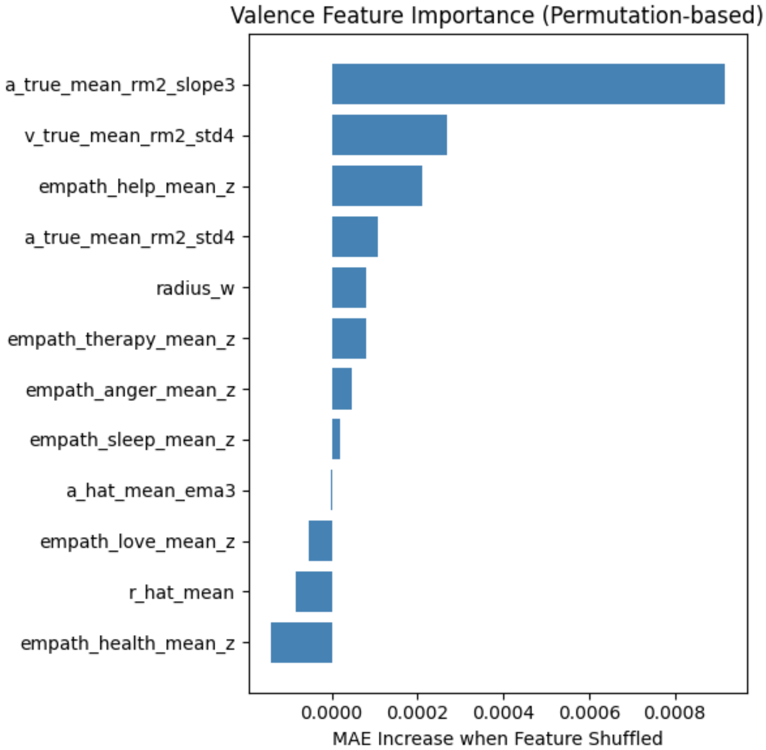

Figure 10: Valence Feature Importance

these patterns:

- v_true_mean_rm2_ema3 accounts for roughly 30% of total attribution mass,
- v_last contributes another 12%,
- and all remaining features individually fall below 5

The interpretability analysis shows that the ARX model forecasts emotional trajectories primarily by extrapolating recent affective momentum, with text-derived emotional signals playing a supplementary but non-dominant role. This explains why fusion improves accuracy (neural and lexicon cues add incremental information) but confirms that short-term forecasting in sparse forum data is largely governed by temporal dynamics rather than isolated linguistic content.

Feature Family Importance To evaluate how different sources of information collectively shape the ARX forecasts, SHAP values were integrated by feature family (Figure 17). Temporal features clearly dominate the model. For valence, they account for approximately 67% of total attribution mass (and 61% for arousal; Appendix A.2). This confirms that recent emotional momentum (captured by rolling means, slopes, and exponential smoothing) contributes much more to short-term forecasting than any single linguistic signal.

Among the auxiliary modalities, Empath categories contribute roughly 21%, reflecting their role in capturing coarse emotional tone. Mental-RoBERTa regressions and Circumplex-derived engineered features each contribute about 6%, indicating that while neural and lexicon-affective cues help the model, they remain secondary to temporal dynamics. Behavioral indicators (post counts, volatility), consistent with ablation analyzes in Section 4.2, exhibit minimal SHAP impact. These results show a strong concordance between model performance (Section 4.2) and global interpretability: temporal structure is the primary driver of next-week emotional forecasts, with text-derived affect serving as complementary context rather than leading signals.

Dependence Patterns To illustrate how individual predictors influence forecasts, Figure 18 presents a dependence graph for the most influential feature, v_true_mean_rm2_ema3. SHAP contributions increase almost linearly as the smoothed valence trend increases, indicating a stable monotonic effect: users showing improving valence in recent weeks are predicted to continue improving. The tight diagonal band reflects consistent interpretation across users, with little inter-user dispersion.

Other features show substantially weaker slopes and lower magnitudes of attribution. For instance, v_last and v_true_mean_rm2_std4 display flatter patterns and a wide spread of SHAP values, consistent with their smaller global importance. Arousal-specific dependence patterns follow the same qualitative structure and are provided in Appendix A.2.

Local Explanation (Representative User-Week) To demonstrate how multi-modal evidence combines for a single prediction, Figure 20 shows a SHAP waterfall plot for one representative user-week (Sample 100). The ARX forecast ($\hat{y}$ 0.031) arises from a small number of interacting contributions. In this example, empath_shame_mean_z yields a positive shift (+0.12), while the dominant temporal feature v_true_mean_rm2_ema3 and the distress-related empath_pain_mean_z exert negative pressure (approximately –0.04 each). Most other predictors contribute less than $\pm$0.02.

This pattern reflects a typical property of the linear ARX architecture: linguistic signals may nudge the forecast upward or downward, but temporal patterns usually determine the baseline movement. Arousal waterfall examples exhibit analogous behavior and are included in Appendix A.2. The local analysis demonstrates that the model's behavior is both interpretable and psychologically plausible—forecasts arise from an additive interplay of temporal momentum, lexical affect, and transformer-inferred cues, rather than from opaque latent states.

4.4 Extended Forecast Diagnostics

4.4.1 Direction Accuracy

To assess whether the ARX model can correctly anticipate week-to-week emotional direction (improvement vs deterioration), we evaluated direction accuracy and macro-F1 across 251 users using user-specific calibrated thresholds. Each user's threshold was set to 0.2 × SD, preventing very small fluctuations near zero from being treated as meaningful directional changes.

Across 251 users, the ARX model achieved:

Macro Direction	Accuracy	F1	Mean thresholds
Valence	0.340	0.286	0.093
Arousal	0.356	0.303	0.091

Table 6: Direction Prediction

These values appear modest, but the task is intrinsically difficult: most users exhibit weekly emotional fluctuations are too small or too irregular to support high directional predictability. This is visible in Figure 11, where most users cluster between 0.20-0.40 accuracy, while a smaller subset of low-volatility users reaches 0.60–0.80. The F1 distribution mirrors this pattern.

The emotional direction prediction is feasible only when users display stable, low-variance trajectories-a finding reinforced by the case studies in Section 4.3.2. These diagnostics motivate the use of volatility-adjusted thresholds and provide guidance for selecting users likely to benefit from early-warning systems.

4.4.2 User Case Studies: Stable vs. Volatile Emotional Patterns

To illustrate how ARX adapts to different behavioral profiles, we examined two representative users drawn from the extremes of the volatility distribution. Volatility is computed as the per-user standard deviation of weekly valence–arousal values and reflects the predictability of a user's emotional trajectory. Corresponding plots are provided in Appendix A.3.

(a) Stable User (ID 1019)

- Weeks: 30,
- Volatility: Valence $\sigma = 0.226$, Arousal $\sigma = 0.266$

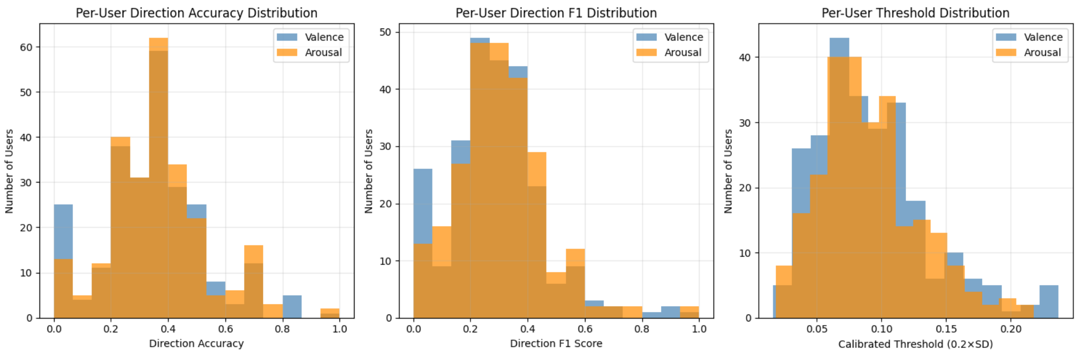

Figure 11: Per-User Direction Distribution

- Valence = 0.647, Arousal = 1.177

This user exhibits gradual emotional drift without abrupt shifts. ARX predictions rely almost entirely on temporal predictors (v_last, v_true_mean_rm2_ema3), and SHAP confirms that these dominate the explanation. Content-based signals play only a minor role. This pattern is characteristic of users whose emotional behavior is well captured by low-order autoregressive dynamics.

(b) Volatile User (ID 4818)

- Weeks: 12,
- Volatility: Valence $\sigma = 1.035$, Arousal $\sigma = 0.960$
- Valence = 3.858, Arousal = 4.199

This user undergoes sharp emotional swings, with large jumps multiple times within a short span. Temporal features alone cannot explain such rapid changes. SHAP analysis shows a much greater contribution from content-based indicators, including empath_shame_mean_z, empath_violence_mean_z, and related distress-language categories. These users require richer linguistic signals to mitigate the difficulty of forecasting high-variance trajectories.

The result shows stable users are modeled primarily through temporal momentum, while volatile users depend more heavily on linguistic cues. These case studies demonstrate that ARX adaptively reallocates feature importance in response to user volatility, validating the decision to combine auto-regressive structure with multi-modal linguistic inputs.

4.4.3 Feature Family Ablation

To quantify how much each feature family contributes to predictive performance, we retrained ARX models after removing one family at a time. Performance degradation (MAE) reflects the importance of the removed group.(Figure12).

The results strongly confirm the SHAP analysis. Removing temporal features causes the largest performance degradation by a wide margin: +0.1935 MAE for valence and +0.2028 MAE for arousal, showing ARX predictions are driven primarily by recent emotional momentum, while linguistic signals offer secondary refinements. Circumplex removal increases MAE slightly (+0.0020 for valence, +0.0018 for arousal), while Empath and Roberta features contribute extremely small and symmetric effects, with MAE changes at or near zero (e.g., Empath: +0.0001, Roberta: –0.0000).

The ranking derived from ablation closely matches the ordering obtained from SHAP importance: (1) Temporal (largest impact), (2) Empath, (3) Roberta, (4) Circumplex.

There are additional analysis in Appendix 4 & 5 due to the content length limitation.

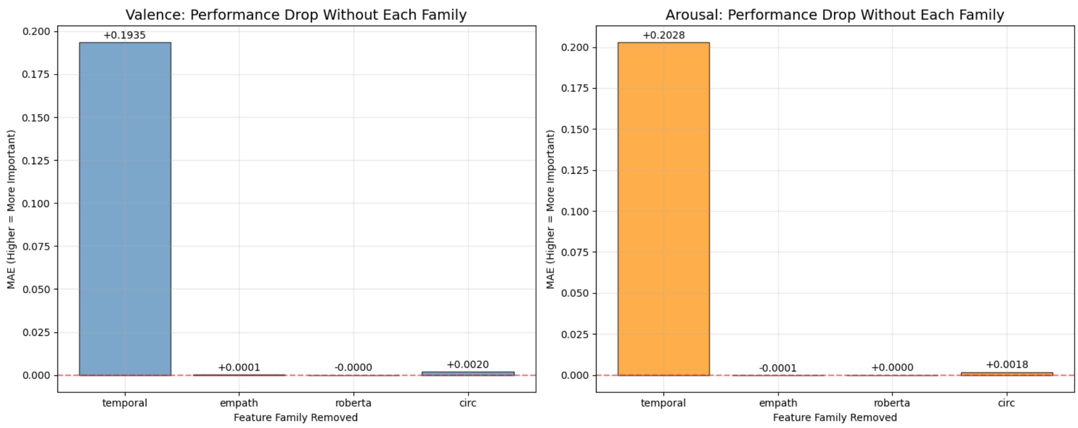

Figure 12: Feature Family Performance Drop Without Each Family

5 Conclusion

This study developed a multi-modal, temporally aware framework for detecting early signs of psychological distress in online mental-health forums. 42,384 posts and comments were used from Beyond Blue to establish a reproducible pipeline integrating psycholinguistic features (Empath), continuous affective representations (Circumplex valence–arousal trajectories), and neural emotion estimates from Mental-RoBERTa. These signals were integrated into weekly user-level trajectories and modeled through a personalized ARX forecaster, with SHAP used to ensure a transparent interpretation of predictions.

Across all components, several consistent findings emerged. First, the emotional landscape of Beyond Blue is overwhelmingly negative and low-energy, with more than 80% of documents concentrated in distressed quadrants of the Circumplex space. Second, Mental-RoBERTa captured a broad affective direction but showed variance compression, highlighting the difficulty of predicting emotion from short, conversational text. Third, ARX substantially outperformed deep recurrent baselines and provided stable, interpretable forecasts. Temporal features (particularly recent valence trends and short-term emotional momentum) were the dominant predictors, while linguistic and neural features provided smaller but complementary gains. SHAP analyzes confirmed these patterns and revealed substantial user-level heterogeneity, with stable users predictable primarily through temporal structure and volatile users requiring richer linguistic cues.

Overall, the results show that early distress forecasting in real-world forums is feasible but inherently constrained by sparse, irregular behavior, and noisy emotional signals. The proposed framework demonstrates how multimodal linguistic, affective, and neural cues can be integrated into an interpretable temporal model that supports week-ahead emotional monitoring. Future work may improve performance through richer inter-actional features (e.g., conversational structure, reply networks), finer-grained LLM regression targets, volatility-aware modeling, and hybrid linear-neural architectures. These advances can enable more reliable, transparent, and clinically aligned early-warning systems for online mental-health settings.

Acknowledgements

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A Appendix

A.1 Symptom Prototype Appendix

To provide interpretable indicators of key mental-health symptom patterns, we constructed five symptom prototype scores: anxiety, depression, PTSD-like arousal, mania, and suicidality. These prototypes represent characteristic emotional "locations" in the valence–arousal space, based on well-established affective science (e.g. anxiety = negative valence + high arousal; depression = negative valence + low arousal). These values

Table 7: Symptom prototypes in the whitened valence–arousal space.

Symptom Type	**v***	**a***	**Notes**
Anxiety	-0.4	+0.6	Negative valence, high arousal
Depression	-0.7	-0.4	Negative valence, low arousal
PTSD	-0.5	+0.5	Intrusive distress, hyperarousal
Mania	+0.4	+0.7	Positive valence, high arousal
Suicidality (agitated)	-0.9	+0.2	Negative valence with agitation
Suicidality (shutdown)	-0.9	-0.2	Negative valence with collapse

are lightly adapted from the empirical centroids in our dataset and the expected affective profile of each symptom.

A.2 SHAP Visualization Appendix

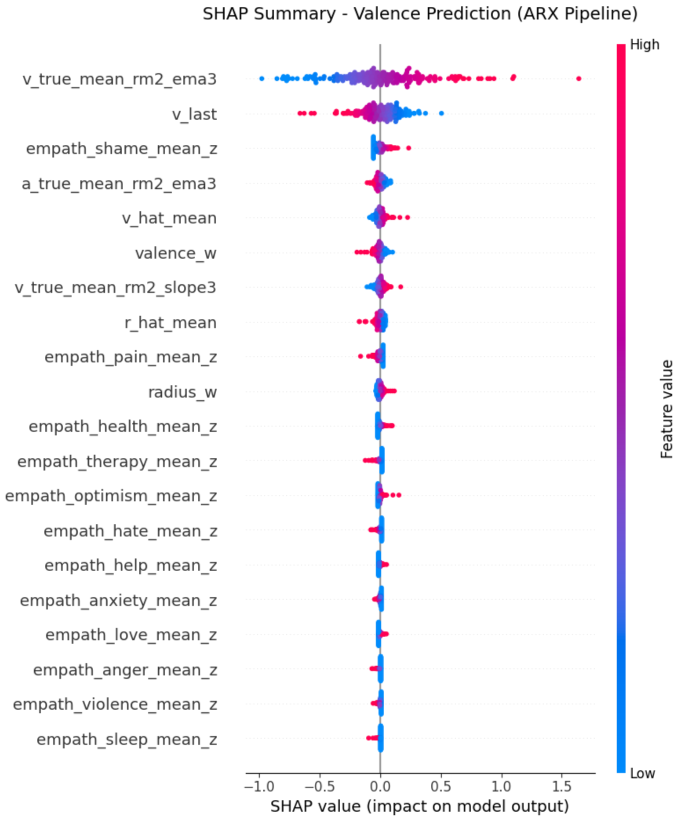

Figure 13: Summary – Valence Prediction

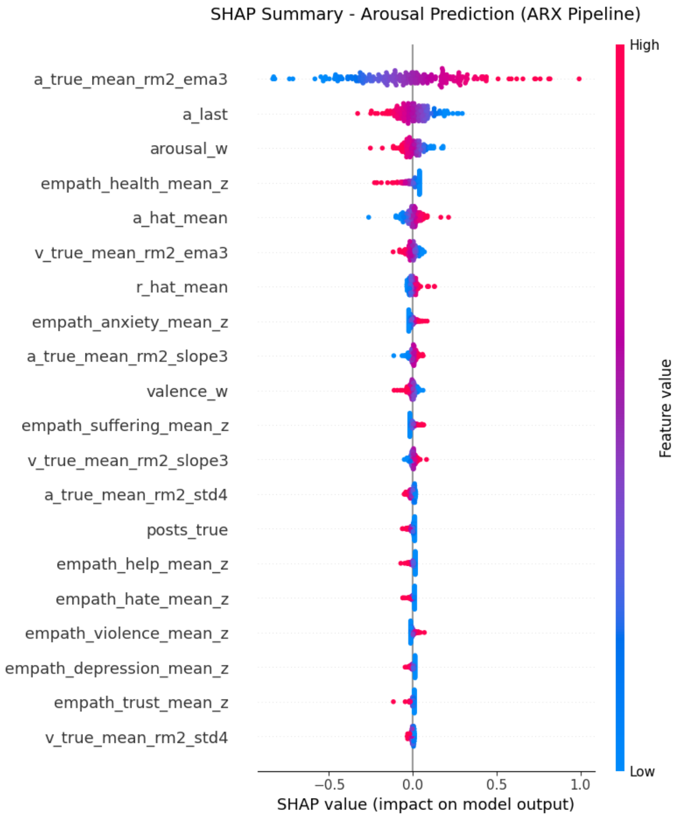

Figure 14: SHAP Summary – Arousal Prediction

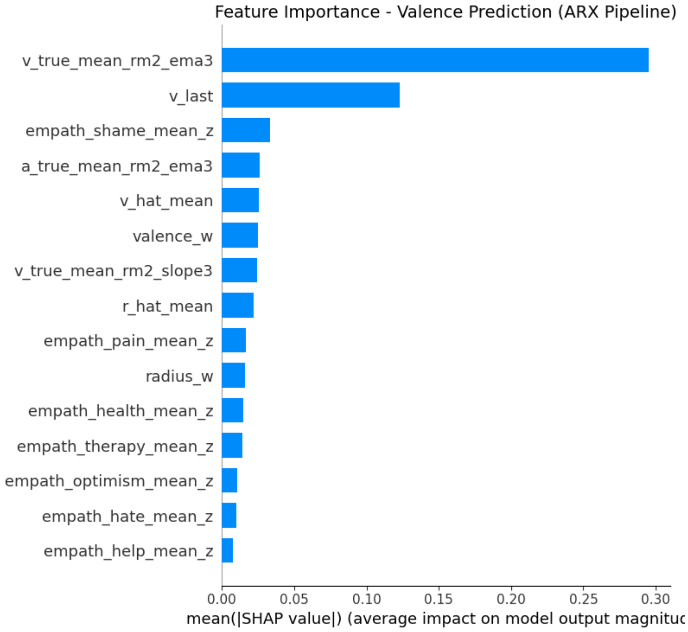

Figure 15: Feature Importance-Valence prediction

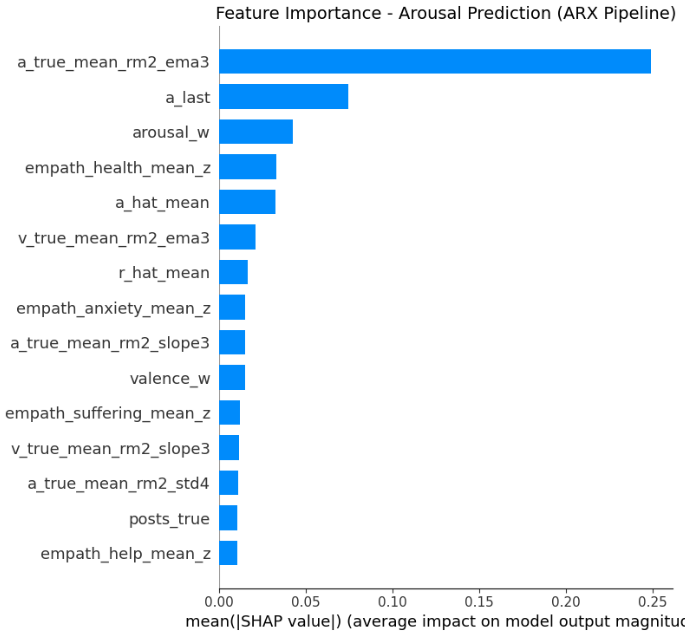

Figure 16: Feature Importance-Arousal prediction

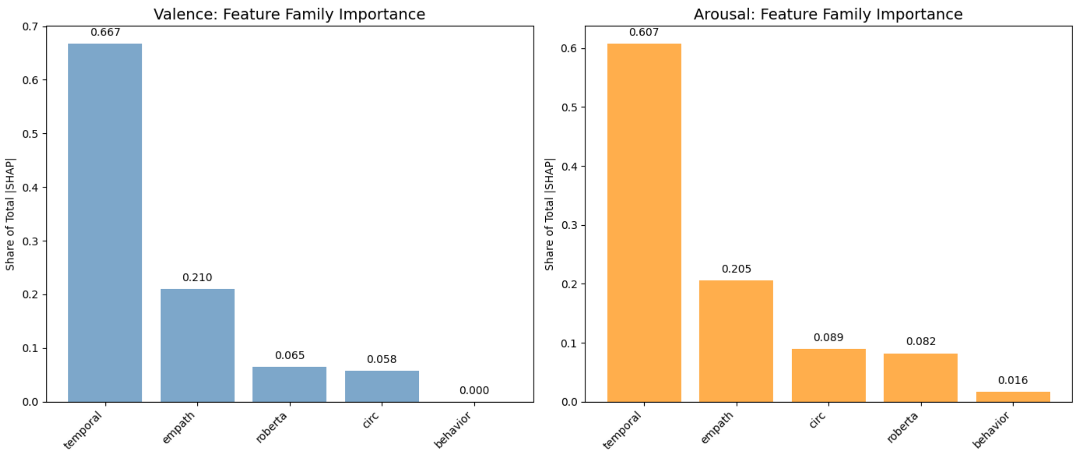

Figure 17: Family Importance

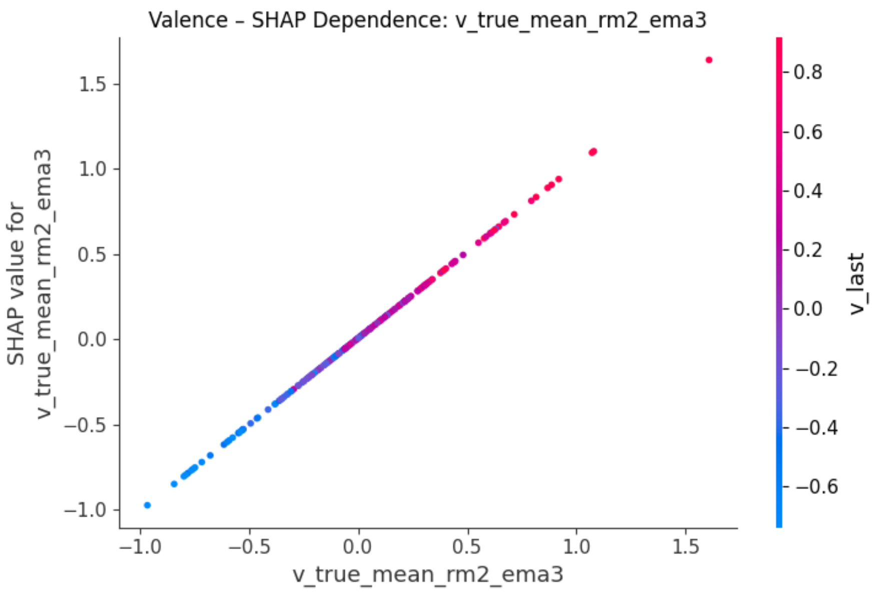

Figure 18: Valence - SHAP Dependence- v_true_mean_rm2_ema3

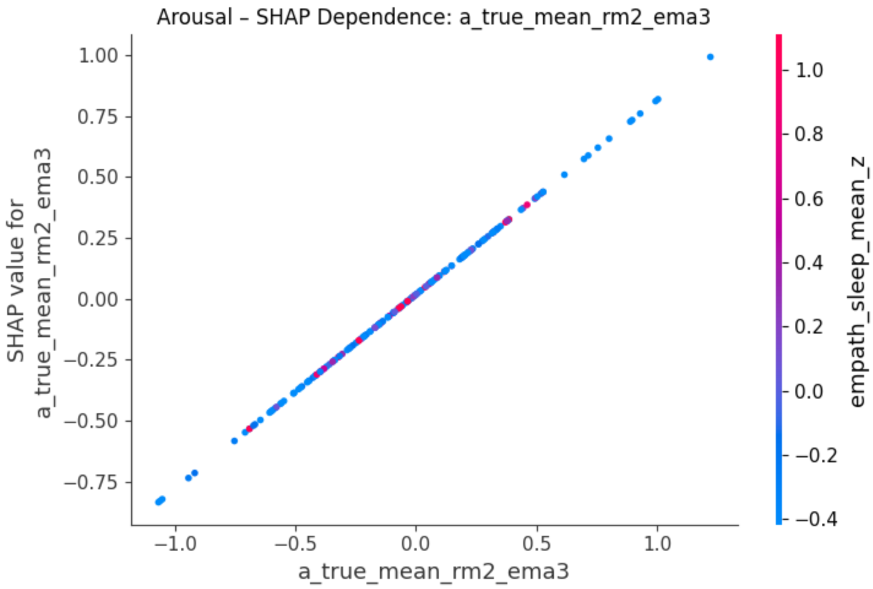

Figure 19: Arousal - SHAP Dependence- a_true_mean_rm2_ema3

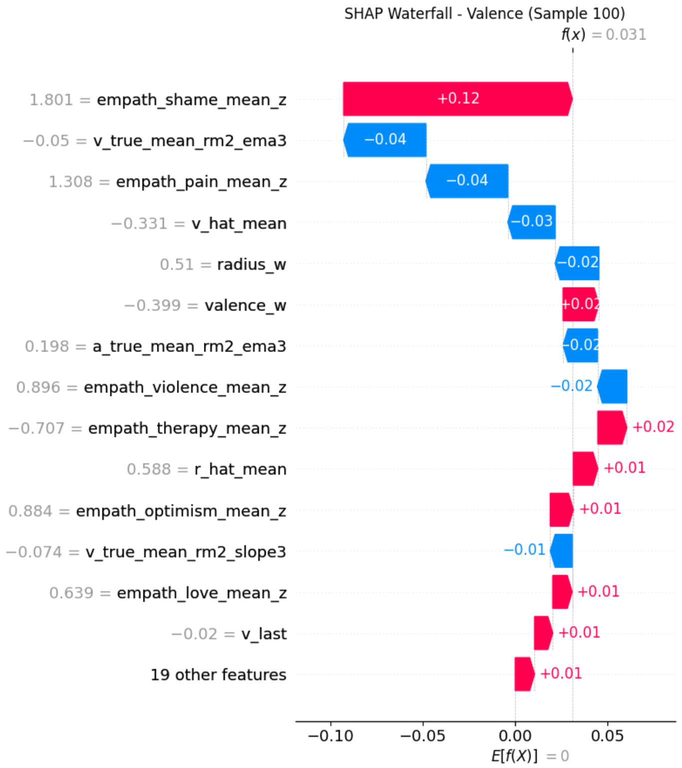

Figure 20: SHAP Waterfall-Valence

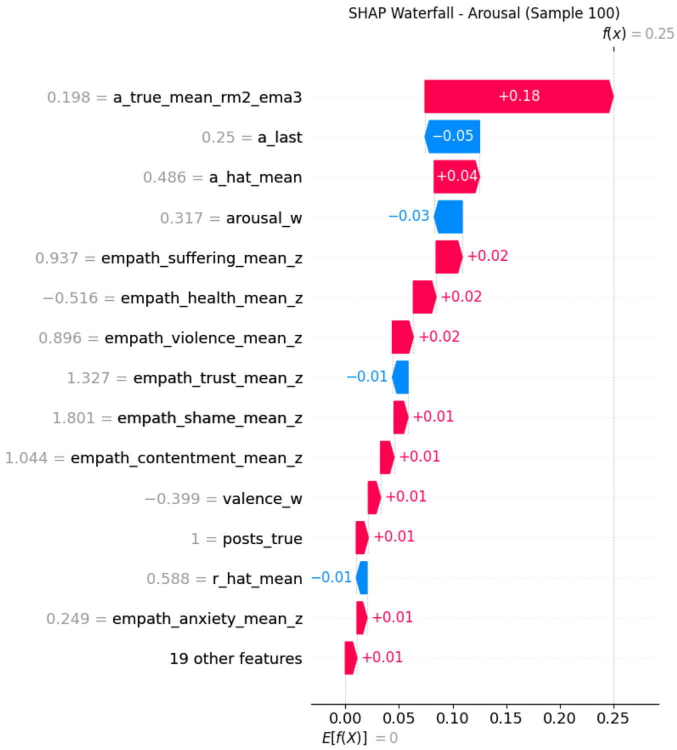

Figure 21: SHAP Waterfall-Arousal

A.3 Extended Forecast Diagnostics Appendix

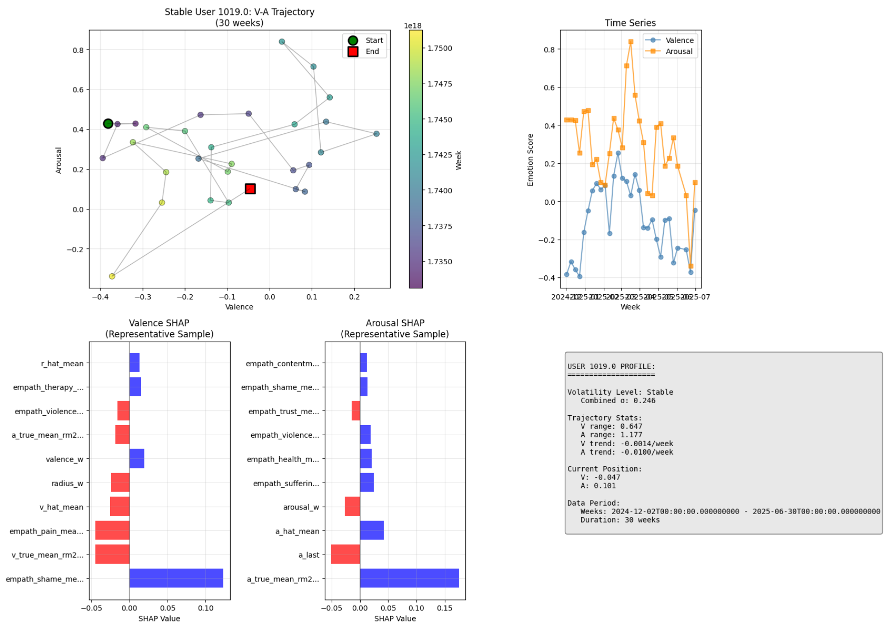

Figure 22: SHAP Waterfall-Arousal

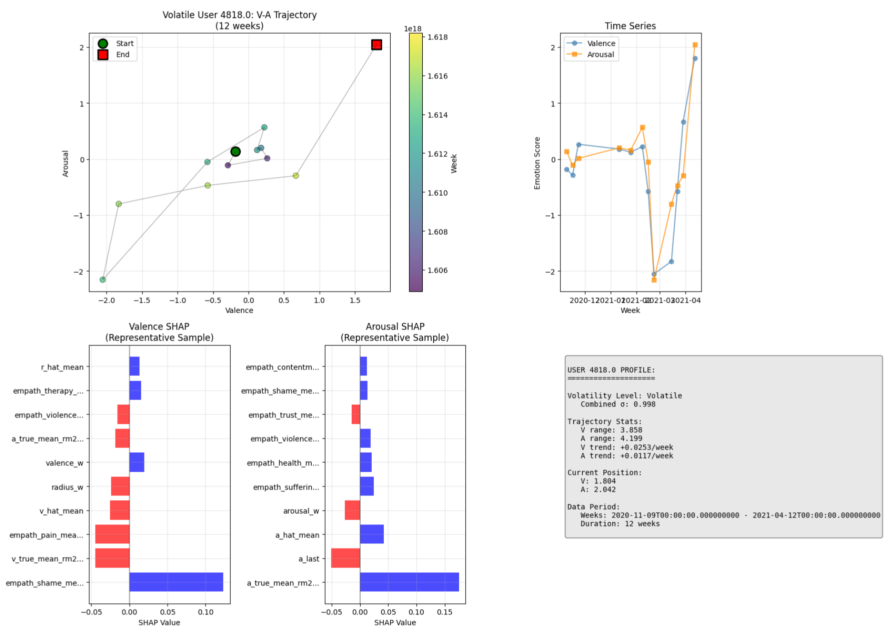

Figure 23: SHAP Waterfall-Arousal

A.4 Uncertainty Quantification: Risk Bands for H-week Forecasts

To assess the reliability of the ARX forecasts, we computed quantile-based prediction bands and conformal prediction intervals. Quantile regression models (10%, 50%, and 90%) were trained for both valence and arousal, and conformal calibration yielded well-calibrated 90% intervals (valence coverage = 0.900, arousal = 0.889). Interval widths were narrow and comparable between the targets (0.23), indicating stable uncertainty estimates.

Risk categories were derived by comparing observed values with the 10–90% bands. Thresholds of approximately 0.15 (low risk) and 0.25 (high risk) divided the test set into low, medium, and high-uncertainty weeks. As expected, forecast error increased monotonically across risk levels: low-risk weeks showed MAE near 0.04, while high-risk weeks exceeded 0.10.

Across all users and weeks, 45.8% of valence–arousal pairs were flagged as high-uncertainty. These reflect periods where emotional trajectories are highly variable or are not adequately supported by linguistic context. These results demonstrate that quantile and conformal methods provide coherent confidence bands and enable user-week–specific risk stratification, supporting downstream monitoring or alerting.(Figure24)

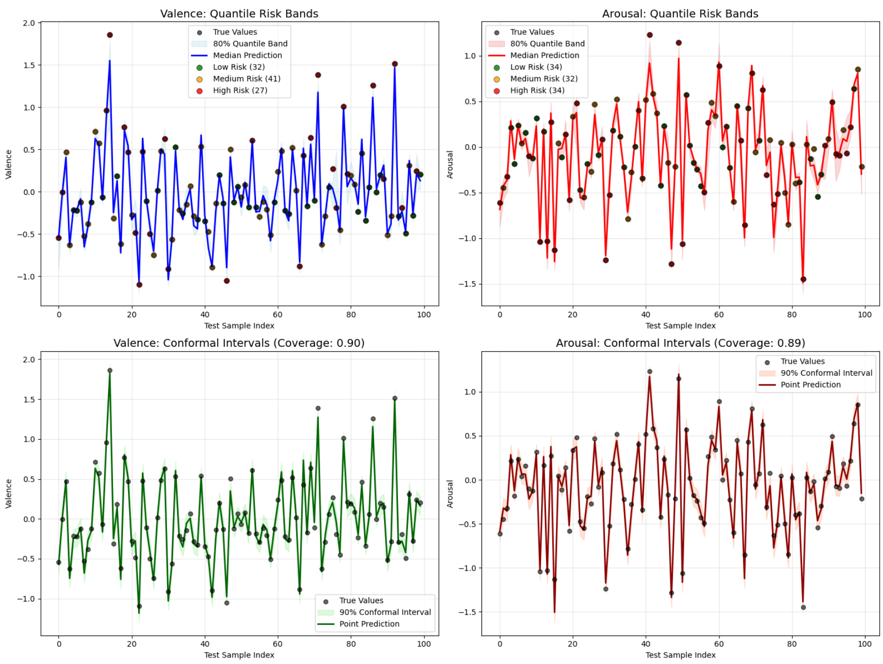

Figure 24: Uncertainty and Extreme-Change Diagnostics

A.5 Extreme-Change Detection: Anomaly Flagging for Hard-to-Predict Jumps

To identify weeks where affective changes were unusually difficult to predict, we analyzed absolute ARX residuals and fitted theoretical distributions to estimate data-driven anomaly thresholds. Valence residuals were best modeled by a lognormal distribution, while arousal residuals were well approximated by a Weibull distribution. Using the 95th-percentile threshold (0.151–0.152), the system flagged 5% of weeks for each target as extreme deviations.

Scatter-plots of predicted vs. actual values show that these anomalies correspond to large, abrupt shifts rather than systematic bias, indicating that emotional "jumps" are driven by sudden changes in user behavior or context rather than model instability. Anomaly rates varied by user (mean = 0.11, range 0–1), suggesting individual differences in volatility and providing potential value for personalized monitoring.

Together, these diagnostics show that the ARX forecasting pipeline can identify both structural uncertainty (wide prediction bands) and behavioral irregularities (extreme residual outliers). These signals may support early-warning systems when combined with clinical thresholds, but are not intended to serve as diagnostic indicators. (Figure25)

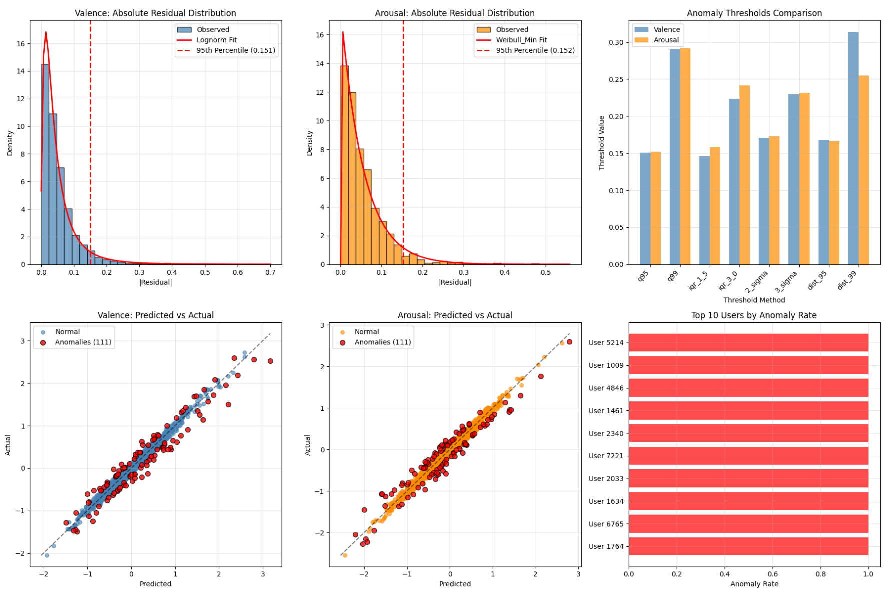

Figure 25: Extreme-Change Detection

B Implementation and Code Availability

[GitHub]

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